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Project Proposal

Team Members & Project Title

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This project groups ~3,000 US counties into a handful of community-health types using K-Means clustering so that our public healthcare system can target different kinds of clusters instead of chasing one county at a time.					

Topic & Summary

Topic of Interest

Most public-health work goes county by county, or it ranks counties from healthiest to worst and sends help to the bottom of the list. A ranking tells you who is struggling. It doesn't tell you what kind of trouble they're in, or what would actually help. A rural county with high diabetes and barely any clinics is a different problem from a wealthy suburb that just scores low on one screening. Call both "unhealthy" and you lose that difference.

We're using the CDC PLACES 2024 dataset, which scores all 3,145 US counties on 40 measures of disease, behavior, and access to care. Instead of ranking, we cluster. We group counties by how similar their full profiles are and let a few types emerge on their own, without deciding upfront what we're after. What comes out is a handful of named types, like high-burden and low-access, or healthy and well-served, plus a map of where each county falls. That lets us treat kinds of places instead of one county at a time, and it flags the counties with the same disease burden but very different access to care.

Summary: 1. Why Is This Topic Important? 2. How Will Data Be Used? 3. Potential Impact

Where you live has a lot to do with how healthy you get to be. Two counties can sit in the same spot on a ranking and still need completely different things. One has high diabetes and barely any clinics. The other is wealthy and just scores low on one screening. Public health tends to work county by county, or it goes after whoever ranks worst, and that misses the pattern underneath. If we can find the natural types of places, we can match help to the type instead of treating every county as its own puzzle!

We will be analyzing the CDC PLACES 2024 dataset. It scores all 3,145 US counties on 40 measures of disease, behavior, and access to care. Instead of ranking, we cluster. We group counties by how similar their full profiles are and let a few types emerge on their own. Those groups will help us answer what kinds of community health are actually out there, and which counties are stuck with the same problem.

The findings can be crucial in curating public health policymaking like bridging the access to healthcare gap. Our model will analyze the way geography affects the prominent diseases across the nation. It would allow government agencies to send the right help to the right kind of places to make sure they do their best with their fixed budgets and counties whose numbers haven't slipped yet get prevention before they do!

Machine Learning Algorithm(s)

Principal Component Analysis: To reduce the dimensionality of 40+ health indicators (example; highly correlated disease rates) to ensure clustering is more stable.	K-means Clustering: To group around 3,000 counties into distinct clusters based on similarities in health outcomes, behaviors, and access measures.	Hierarchical Clustering: to generate a dendrogram that helps confirm if the number of clusters (k) found by K-Means is natural or forced

Research Question

Can we cluster diseases through geographic, disease, and other parameters to identify patterns, and if so, what are they? among ~3000 US counties to government agencies to take action on a systematic level!

Dataset(s)

- **Name:** US CDC PLACES County Health Data 2024
- **Source:** Kaggle
- **Link:** https://www.kaggle.com/datasets/aliafzal9323/us-cdc-places-county-health-data-2024

Training & Testing Considerations

Our dataset covers all 3143 counties in the United States, so our data is representative of different geographic locations. Additionally, our machine learning model of K-Means Clustering can take any different number of parameters to identify the width of clusters to see if there are certain diseases that are more prominent within different geographies. Finding these differences and similarities can help identify potential trends as well as help us infer underlying issues such as a lack of governance or a lack of clinics in the counties.

Sources of Bias

Model Bias PLACES values are estimated from a survey plus demographics, not counted, so similar-looking counties get similar numbers by default	**Aggregation Bias** One county-wide average hides the rich and poor neighborhoods inside it	**Sampling Bias** It's a self-report phone survey, so it misses people without phones, non-English speakers, and undercounts stigmatized conditions

Mitigation Strategies

To mitigate model bias, we plan to use the confidence limit columns to flag shaky estimates and report them	To mitigate aggregation bias, we would state that types describe counties, not individuals, and never act on a single county number alone	To mitigate sampling bias, we would treat results as approximate and note which groups the survey underrepresents

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Citations

Bowser, D. M., Mauricio, K., Ruscitti, B. A., & Crown, W. H. (2024). American clusters: Using machine learning to understand health and health care disparities in the United States. *Health Affairs Scholar, 2*(3), qxae017. https://doi.org/10.1093/haschl/qxae017	Khan, S. S., Krefman, A. E., McCabe, M. E., Petito, L. C., Yang, X., Kershaw, K. N., Pool, L. R., & Allen, N. B. (2022). Association between county-level risk groups and COVID-19 outcomes in the United States: A socioecological study. *BMC Public Health, 22*, Article 81. https://doi.org/10.1186/s12889-021-12469-y	Greenlund, K. J., Lu, H., Wang, Y., Matthews, K. A., LeClercq, J. M., Lee, B., & Carlson, S. A. (2022). PLACES: Local data for better health. *Preventing Chronic Disease, 19*, 210459. https://doi.org/10.5888/pcd19.210459